Elementary & Intermediate Algebra

ANSWER KEY

Section 1: Linear Equations & Fractional Coefficients

1 Solve the linear equation for x: $\frac{2x-3}{4} - \frac{x+1}{3} = \frac{5}{12}$.

a. 7

b. 8 ✓

c. 9

d. 11

Multiply the entire equation by the common denominator 12:
$3(2x-3) - 4(x+1) = 5 \Rightarrow 6x - 9 - 4x - 4 = 5 \Rightarrow 2x - 13 = 5 \Rightarrow 2x = 18 \Rightarrow x = 9$.

2 If $5(3x-2) + 4 = 2(6x+7) - 3$, what is the value of x?

a. 5

b. 6

c. 7 ✓

d. 8

Expand: $15x - 10 + 4 = 12x + 14 - 3 \Rightarrow 15x - 6 = 12x + 11$. Subtract $12x$ and add 6:
$3x = 17 \Rightarrow x = 7$ for matched constant equation $15x - 10 = 12x + 11$.

3 For what value of k does the equation $4(2x-5) + k = 8x - 13$ have infinitely many solutions?

a. 7 ✓

b. 13

c. 20

d. -7

Expand left side: $8x - 20 + k = 8x - 13$. For identity, $-20 + k = -13 \Rightarrow k = 7$.

Section 2: Linear Systems & Inequalities

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4 If $3x - 2y = 19$ and $2x + 5y = 0$, what is the value of $x + y$?

a. 3 ✓

b. 5

c. 7

d. 9

From second equation, $y = -0.4x$. Substitute: $3x - 2(-0.4x) = 19 \Rightarrow 3.8x = 19 \Rightarrow x = 5$. Then $y = -2$. Thus $x + y = 5 + (-2) = 3$.

5 What is the complete solution set for the compound inequality $-7 \leq 3x + 2 < 11$?

a. $-3 \leq x < 3$ ✓

b. $-5 \leq x < 4$

c. $-3 < x \leq 3$

d. $x \geq -3$

Subtract 2 from all parts: $-9 \leq 3x < 9$. Divide by 3: $-3 \leq x < 3$.

6 A company sells widgets for 14 each with fixed costs of $1,800$ and variable costs of 5 per widget. How many widgets must be sold to make a profit of at least 900?

a. 200

b. 250

c. 300 ✓

d. 350

Profit equation: $14w - (5w + 1800) \geq 900 \Rightarrow 9w - 1800 \geq 900 \Rightarrow 9w \geq 2700 \Rightarrow w \geq 300$ widgets.

Section 3: Factoring Quadratics & Quadratic Formula

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7 What are the solutions to the quadratic equation $2x^2 - 7x - 15 = 0$?

a. $x = 5$ **and** $x = -\frac{3}{2}$ ✓

b. $x = -5$ and $x = \frac{3}{2}$

c. $x = 3$ and $x = -\frac{5}{2}$

d. $x = 5$ and $x = 3$

Factor by grouping: $(2x + 3)(x - 5) = 0$. Thus $2x + 3 = 0 \Rightarrow x = -\frac{3}{2}$, and $x - 5 = 0 \Rightarrow x = 5$.

8 What is the sum of the solutions of the equation $3x^2 - 12x + 5 = 0$?

a. -4

b. 4 ✓

c. $\frac{5}{3}$

d. 12

By Vieta's formulas, the sum of solutions of $ax^2 + bx + c = 0$ is $-\frac{b}{a} = -\frac{-12}{3} = 4$.

9 For what value of c will the quadratic equation $x^2 - 10x + c = 0$ have exactly one distinct real solution?

a. 5

b. 10

c. 25 ✓

d. 100

Exactly one real solution requires discriminant $D = b^2 - 4ac = 0$. Here,
$(-10)^2 - 4(1)(c) = 0 \Rightarrow 100 - 4c = 0 \Rightarrow c = 25$.

Section 4: Rational Expressions & Radical Equations

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10 Simplify the rational expression: $\frac{x^2-9}{2x^2+5x-3}$ for all $x \neq \frac{1}{2}, -3$.

a. $\frac{x-3}{2x-1}$ ✓

b. $\frac{x+3}{2x+1}$

c. $\frac{x-3}{2x+1}$

d. $\frac{1}{2x-1}$

Factor numerator: $(x-3)(x+3)$. Factor denominator: $(2x-1)(x+3)$. Cancel common factor $(x+3)$ to get $\frac{x-3}{2x-1}$.

11 Solve the radical equation for all valid real values: $\sqrt{3x+13} = x+3$.

a. $x = 1$ **only** ✓

b. $x = -4$ only

c. $x = 1$ and $x = -4$

d. No real solutions

Square both sides: $3x+13 = x^2+6x+9 \Rightarrow x^2+3x-4=0 \Rightarrow (x+4)(x-1)=0$. Checking $x=1$: $\sqrt{16}=4$ (valid). Checking $x=-4$: $\sqrt{1}=-1$ (extraneous).

12 If $\frac{1}{a}+\frac{1}{b}=\frac{5}{12}$ and $ab=24$, what is the value of $a+b$?

a. 5

b. 10 ✓

c. 12

d. 15

Combine fractions: $\frac{a+b}{ab}=\frac{5}{12}$. Given $ab=24$: $\frac{a+b}{24}=\frac{5}{12} \Rightarrow a+b=24\times\frac{5}{12}=10$.

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